Epilepsy is a chronic neurological disorder characterised by an enduring predisposition to generate unprovoked seizures. A seizure is a transient occurrence of signs or symptoms due to abnormal excessive or synchronous neuronal activity in the brain.

Focal seizures originate within networks limited to one hemisphere. Temporal lobe epilepsy is the most common focal epilepsy in adults and frequently involves the hippocampus, amygdala and parahippocampal gyrus. Frontal lobe seizures are often brief, occur from sleep, and show prominent motor features such as hypermotor or tonic posturing.

On scalp EEG, interictal epileptiform discharges include spikes (20-70 ms) and sharp waves (70-200 ms), often followed by a slow wave. Generalised 3 Hz spike-and-wave discharges are the electrographic hallmark of childhood absence epilepsy. Rhythmic evolving activity in the theta or alpha range may mark the onset of a focal electrographic seizure.

Diagnosis relies on clinical history, EEG and neuroimaging. MRI with epilepsy protocol is recommended to identify structural causes such as hippocampal sclerosis, focal cortical dysplasia, tumours and vascular malformations. Prolonged video-EEG monitoring helps localise the seizure onset zone.

First-line treatment of focal seizures includes carbamazepine, lamotrigine and levetiracetam. Generalised tonic-clonic seizures are commonly treated with valproate, lamotrigine or levetiracetam. Drug-resistant epilepsy, defined as failure of two appropriately chosen and tolerated antiseizure medications, should prompt referral for surgical evaluation.

Status epilepticus is a neurological emergency defined by a seizure lasting more than five minutes or recurrent seizures without recovery. Initial management uses intravenous or intramuscular benzodiazepines, followed by a second-line antiseizure medication such as levetiracetam or fosphenytoin.

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